

MEHRAB HASAN, MS

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SUMMARY

- 3+ years of experience in Advanced Manufacturing and Shape Memory Polymer, Machine Learning assisted prediction model research. Proficient in Experimental Setup Design, Simulation (ANSYS,SolidWorks), Data Analysis (Python, MATLAB)

TECHNICAL SKILLS

- Material Characterization: Raman spectroscopy, SEM, EIS, Cyclic Voltammetry
- Material Synthesis: Hydrothermal synthesis, CVD, Drop-casting
- Programming: MATLAB, Python (PyBaMM)
- Battery Fabrication & Testing: MACCOR, NEWARE, Glovebox, Gamry, Coin cell fabrication
- Simulation & Design: ANSYS,SolidWorks, Fusion360, AutoCAD
- Data Analysis & Visualization: Origin, Minitab, Google Data Studio, Power BI, Microsoft Excel
- Additive Manufacturing: DIW, FDM, DLP

RESEARCH EXPERIENCE

University of Oklahoma

Norman, Oklahoma

Graduate Research Assistant

June 2024 to Present

- Developed a machine learning assisted shape recovery prediction model for shape memory epoxy polymer composite.
- Designed experimental setup and performed disassembly of Li-ion pouch cells to fabricate coin cells with harvested electrodes.
- Developed physics-based degradation models for Li-ion batteries with graphite-silicon composite electrodes.
- Implemented a Model Predictive Controller for Li-metal batteries considering temperature, voltage, and electrolyte concentration as constraints.

University of North Texas

Denton, Texas

Graduate Research Assistant

June 2023 to May 2024

- Developed Iron-CO₂ batteries and improved battery specific capacity by 900% with the application of 2D TMDs catalysts (MoS₂, MoWS₂)
- Developed Hydrothermal synthesis procedures of MoS₂ for catalyst application.
- Performed long-cycle charge discharge test, EIS, CV tests to analyze electrochemical cell performance.
- Developed additively manufactured Zn-V₂O₅ cells with different electrode thickness layers to achieve enhanced cell capacity.
- Developed thin film PVDF-MoSe₂ piezoelectric sensors to gain boosted performance.
- Analyzed experimental data to make data-driven decisions and project optimization.
- Worked as Lab Inventory Manager , performed regular maintenance and troubleshooting of Glovebox.

PUBLICATIONS

- **Hasan M**, Ghamarian I, Liu Y. Machine Learning Assisted Prediction of Recovered Angle of Carbon Fiber Reinforced Shape Memory Epoxy Carbon Composites. Macromol Rapid Commun. 2026 Jan20:e00962.doi:10.1002/marc.202500962.
- **Hasan, M.**; Liu, Y. 4D-Printed Liquid Crystal Elastomers: Printing Strategies, Actuation Mechanisms, and Emerging Applications. J. Compos. Sci. 2025, 9, 633. <https://doi.org/10.3390/jcs9110633>

EDUCATION

University of Oklahoma

Norman, Oklahoma

PhD in Aerospace and Mechanical Engineering Current GPA 4.00/4.00

June 2024 to Present

University of North Texas

Denton, Texas

MS in Material Science and Engineering GPA 3.78/4.00

August 2022 to May 2024

Bangladesh University of Engineering and Technology

Dhaka,Bangladesh

BS in Naval Architecture and Marine Engineering GPA 3.51/4.00 Position:6th

February 2016 to February 2021

TEACHING AND MENTORING EXPERIENCE

Univeristy of Oklahoma

Norman, Oklahoma

Graduate Teaching Assistant

August 2025 to Present

- Working as a Mentor, Course coordinator for Undergraduate level classes of 130 Students (Aerodynamics, Solid Mechanics).Working as SolidWorks, Ansys Lab instructor for 200 students.

University of North Texas

Denton, Texas

Graduate Teaching Assistant

August 2022 to May 2023

- Mentored 5 Senior Design student groups and guided them to develop their prototypes and experimental setup for the projects.Held problem solving sessions for the Thermodynamics course.

AWARDS

- President's Council Scholarship, University of North Texas (2022-2023)
- BSTF Travel Grant, Bangladesh Sweden Trust Fund (2022-2023)

Studying in the US under F1 visa, authorized to work under OPT/CPT

